



9833 - Journey to the Edge of WD 1145+017's Circumstellar Disk

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Erika Le Bourdais (PI) (CSA Member)	Universite de Montreal
Dr. Patrick Dufour (CoI) (CSA Member)	Universite de Montreal
Dr. Siyi Xu (CoI) (US Admin CoI)	NOIRLab - Gemini North (HI)
Dr. Andrew Swan (CoI) (ESA Member)	University of Warwick
Sydney Jenkins (CoI)	Massachusetts Institute of Technology
Dr. Loic Albert (CoI) (CSA Member)	Universite de Montreal

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	MIRI Imaging	MIRI Imaging	(1) WD1145+017
	2	MIRI LRS	MIRI Low Resolution Spectroscopy	(1) WD1145+017
	3	NIRSpec	NIRSpec Fixed Slit Spectroscopy	(1) WD1145+017

ABSTRACT

WD 1145+017, a polluted white dwarf (WD) with a complex debris disk, offers a prime window into the final stages of planetary systems. This star displays an unprecedented number of metals in its photosphere and its circumstellar gas, and also exhibits transits from an actively disintegrating planetesimal. All combined, WD 1145+017's replenishing circumstellar disk provides a rich environment to study the pollution process of white dwarfs. Despite being one of the most studied WDs, our understanding of its disk in the near- and mid-infrared remains limited.

We propose a comprehensive JWST observing program using MIRI and NIRSpec to fully characterize the thermal and radial structure and determine the mineralogy of WD 1145+017's dust disk. By analyzing the 10 microns silicate feature and mapping the disk's configuration, we aim to provide

JWST Proposal 9833 (Created: Friday, March 13, 2026, 1:00:26PM Eastern Standard Time) - Overview

the first complete picture of the pollution process of white dwarfs from the outer edge of the circumstellar disk to the star's photosphere. This study will improve our understanding of white dwarf disks while providing crucial insights into exoplanetary composition and evolution, bridging the gap between theoretical models and observational data.

OBSERVING DESCRIPTION

We use NIRSpect low resolution spectroscopy to constrain the continuum of the disk at shorter wavelengths. MIRI LRS observations are added to characterize the disk's mineralogy through well recognized features, namely the 10 microns silicate feature. Finally, we extend the wavelength coverage to 20 microns to obtain the remaining tail of the disk continuum as well as probe potential cooler dust contributions.

Proposal 9833 - Targets - Journey to the Edge of WD 1145+017's Circumstellar Disk

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	WD1145+017	RA: 11 48 33.6297 (177.1401238d)	Proper Motion RA: -43.664 mas/yr	
			Dec: +01 28 59.42 (1.48317d)	Proper Motion Dec: -4.027000045425666 mas/yr	
			Equinox: J2000	Parallax: 0.0068534"	
				Epoch of Position: 2000	
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Star</i> <i>Description=[White dwarfs]</i> <i>Extended=NO</i>					

Proposal 9833 - Observation 1 - Journey to the Edge of WD 1145+017's Circumstellar Disk

Observation	Proposal 9833, Observation 1: MIRI Imaging										Fri Mar 13 18:00:26 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	WD1145+017	RA: 11 48 33.6297 (177.1401238d) Dec: +01 28 59.42 (1.48317d) Equinox: J2000			Proper Motion RA: -43.664 mas/yr Proper Motion Dec: -4.027000045425666 mas/yr Parallax: 0.0068534" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star Description=[White dwarfs] Extended=NO										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	5						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1500W	FASTR1	30	1	1	Dither 1	5	5	416.256	260752
	2	F1800W	FASTR1	40	3	1	Dither 1	5	15	1692.774	260752
	3	F2100W	FASTR1	15	10	1	Dither 1	5	50	2206.157	260752

Proposal 9833 - Observation 1 - Journey to the Edge of WD 1145+017's Circumstellar Disk

Special Requirements	Background Limited. Background no more than 10th percentile above minimum Sequence Observations 1, 2, 3, Non-interruptible
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Proposal 9833 - Observation 2 - Journey to the Edge of WD 1145+017's Circumstellar Disk

Observation	Proposal 9833, Observation 2: MIRI LRS								Fri Mar 13 18:00:26 GMT 2026	
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous			
	(1)	WD1145+017	RA: 11 48 33.6297 (177.1401238d) Dec: +01 28 59.42 (1.48317d) Equinox: J2000		Proper Motion RA: -43.664 mas/yr Proper Motion Dec: -4.027000045425666 mas/yr Parallax: 0.0068534" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.									
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.									
	Category=Star Description=[White dwarfs] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	1 WD1145+017	F560W	FAST	10	1	1	27.75	260752	
Template	Subarray				Obtain Verification Image?					
	FULL				true					
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	25	1	1	1	1	69.376		F560W

Proposal 9833 - Observation 2 - Journey to the Edge of WD 1145+017's Circumstellar Disk

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	220	10	20	1	2	12260.127	
Special Requirements	Sequence Observations 1, 2, 3, Non-interruptible								

Proposal 9833 - Observation 3 - Journey to the Edge of WD 1145+017's Circumstellar Disk

Observation	Proposal 9833, Observation 3: NIRSpec										Fri Mar 13 18:00:26 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	WD1145+017	RA: 11 48 33.6297 (177.1401238d) Dec: +01 28 59.42 (1.48317d) Equinox: J2000			Proper Motion RA: -43.664 mas/yr Proper Motion Dec: -4.027000045425666 mas/yr Parallax: 0.0068534" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[White dwarfs] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	WATA	FULL	F110W	NRSRAPID	3	1	1	42.947	260752	
Template	HFF Readout Mode			Slit			Subarray					
	false			S200A1			ALLSLITS					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	S200A1	NRSRAPID	16	2	1	NONE	3	6	560.511	

Proposal 9833 - Observation 3 - Journey to the Edge of WD 1145+017's Circumstellar Disk

Special Requirements	Sequence Observations 1, 2, 3, Non-interruptible
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